

4 - Theory - HW

Decrypt the following Cipher Using the Hill Cipher

1. Cipher text: HLDTRZHXWACMAG

Key: $\begin{bmatrix} 3 & 2 \\ 5 & 7 \end{bmatrix}$ use (A=0, B=1, ..., Z=25)

Find the plain text

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
16	17	18	19	20	21	22	23	24	25						
Q	R	S	T	U	V	W	X	Y	Z						

$$\begin{bmatrix} 3 & 2 \\ 5 & 7 \end{bmatrix} \rightarrow (3 \times 7) - (2 \times 5) = 21 - 10 = 11 \pmod{26}$$

$$11x = 1 \pmod{26}$$

$$11 \times 19 = 209$$

$$209 = 26(8) + 1$$

$$209 \pmod{26} = 1$$

$$11^{-1} = 19 \pmod{26}$$

$$K^{-1} = \det^{-1} = \begin{bmatrix} 7 & -2 \\ -5 & 3 \end{bmatrix} \pmod{26}$$

$$K^{-1} = 19 \begin{bmatrix} 7 & 24 \\ 21 & 3 \end{bmatrix} = \begin{bmatrix} 133 & 456 \\ 399 & 57 \end{bmatrix} \pmod{26} = \begin{bmatrix} 3 & 14 \\ 9 & 15 \end{bmatrix}$$

$$HL = [7, 11] = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 7 \\ 11 \end{bmatrix} = \begin{bmatrix} 21 + 154 \\ 63 + 55 \end{bmatrix} = \begin{bmatrix} 175 \\ 118 \end{bmatrix} = \begin{bmatrix} 19 \\ 14 \end{bmatrix} = T O$$

$$DT = [3, 19] = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 3 \\ 19 \end{bmatrix} = \begin{bmatrix} 9 + 266 \\ 27 + 95 \end{bmatrix} = \begin{bmatrix} 275 \\ 120 \end{bmatrix} = \begin{bmatrix} 15 \\ 18 \end{bmatrix} = P S$$

$$AJ = [16, 8] = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 16 \\ 8 \end{bmatrix} = \begin{bmatrix} 48 + 112 \\ 144 + 40 \end{bmatrix} = \begin{bmatrix} 160 \\ 184 \end{bmatrix} = \begin{bmatrix} 4 \\ 2 \end{bmatrix} = E C$$

$$HJ = [7, 9] = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 7 \\ 9 \end{bmatrix} = \begin{bmatrix} 21 + 126 \\ 63 + 45 \end{bmatrix} = \begin{bmatrix} 147 \\ 108 \end{bmatrix} = \begin{bmatrix} 17 \\ 4 \end{bmatrix} = R E$$

$$DX = [3, 23] = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 3 \\ 23 \end{bmatrix} = \begin{bmatrix} 9 + 322 \\ 27 + 115 \end{bmatrix} = \begin{bmatrix} 331 \\ 142 \end{bmatrix} = \begin{bmatrix} 19 \\ 12 \end{bmatrix} = T M$$

$$WA[22, 16] = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 22 \\ 16 \end{bmatrix} = \begin{bmatrix} 66 + 224 \\ 198 + 80 \end{bmatrix} = \begin{bmatrix} 290 \\ 278 \end{bmatrix} = \begin{bmatrix} 4 \\ 18 \end{bmatrix} = F S$$

$$CM[12, 12] = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 12 \\ 12 \end{bmatrix} = \begin{bmatrix} 6 + 168 \\ 18 + 60 \end{bmatrix} = \begin{bmatrix} 174 \\ 78 \end{bmatrix} = \begin{bmatrix} 18 \\ 0 \end{bmatrix} = S A$$

$$AG[0, 6] = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 0 \\ 6 \end{bmatrix} = \begin{bmatrix} 0 + 84 \\ 0 + 30 \end{bmatrix} = \begin{bmatrix} 84 \\ 30 \end{bmatrix} = \begin{bmatrix} 6 \\ 4 \end{bmatrix} = G E$$

Plaintext: TOP SECRET MESSAGE

2. Consider the hill Cipher Key Matrix

$$\begin{bmatrix} 24 & 19 \\ 5 & 14 \end{bmatrix}$$

Cipher text: RXADY

find the plain text for the above cipher text

→ decryption: $K^{-1} C \pmod{26}$

$$\det(K) = (24 \times 14) - (19 \times 5) = 336 - 95 = 241 = 7 \pmod{26}$$

$$\text{inverse} = 7x = 1 \pmod{26}$$

$$7 \times 15 = 105$$

$$105 \pmod{26} = 1$$

$$7^{-1} = 15$$

$$K^{-1} = 15 \begin{bmatrix} 14 & -19 \\ -5 & 24 \end{bmatrix} = 15 \begin{bmatrix} 14 & 7 \\ 21 & 24 \end{bmatrix} = \begin{bmatrix} 210 & 105 \\ 315 & 360 \end{bmatrix} \pmod{26}$$

$$K^{-1} = \begin{bmatrix} 2 & 1 \\ 3 & 22 \end{bmatrix}$$

$$RX = [17, 23] = \begin{bmatrix} 2 & 1 \\ 3 & 22 \end{bmatrix} \begin{bmatrix} 17 \\ 23 \end{bmatrix} = \begin{bmatrix} 34 + 23 \\ 51 + 506 \end{bmatrix} = \begin{bmatrix} 57 \\ 557 \end{bmatrix} = \begin{bmatrix} 5 \\ 11 \end{bmatrix} = F L$$

$$AO = [0, 14] = \begin{bmatrix} 2 & 1 \\ 3 & 22 \end{bmatrix} \begin{bmatrix} 0 \\ 14 \end{bmatrix} = \begin{bmatrix} 14 \\ 308 \end{bmatrix} = \begin{bmatrix} 14 \\ 22 \end{bmatrix} = O W$$

$$DY = [3, 24] = \begin{bmatrix} 2 & 1 \\ 3 & 22 \end{bmatrix} \begin{bmatrix} 3 \\ 24 \end{bmatrix} = \begin{bmatrix} 6 + 24 \\ 9 + 528 \end{bmatrix} = \begin{bmatrix} 30 \\ 537 \end{bmatrix} \pmod{26} = \begin{bmatrix} 4 \\ 17 \end{bmatrix} = F R$$

Plaintext: FLOWER

3.

Confusion vs diffusion

Analyse the following classical cipher and Confusion diffusion Complete the table & justify your classification based on how each cipher transforms the plaintext

→ Cipher does it change the character/symbol Value? does it change the character position? Confusion justification

Shift Cipher	Yes	No	Confusion	Substituted via mathematical Shift (P+K); character order is unaltered
multiplicative Cipher	Yes	No	Confusion	Substituted via modular multiplication (P+K); character order is unaltered
affine Cipher	Yes	No	Confusion	Combines multiplicative & additive Shift (P+K) ((a-P+K) mod 26)); position remains unchanged
transposition Cipher	No	Yes	diffusion	Character retain original values but their position is altered across the text

Relationship b/w Substitution & Confusion

In a Substitution Cipher, the position remains the same but the actual character value changes

A → O

∴ the relationship b/w the plaintext characters and cipher text character is hidden. Hence Substitution mainly provides Confusion

Relationship b/w transposition and diffusion

In transposition, the character itself doesn't change

ABCDEF → EADBCF

∴ The characters are moved to different positions. This spreads the information of the plaintext throughout the ciphertext. Hence transposition mainly provides Diffusion

4. Transposition

Encryption:

encrypt the message "the answer is pointing north direction" using

→ Simple

with Key = 4

→ Writing is done by rows into 4 columns

T H E A

R R O W

I D P O

I N T I

N G N O

A T H D

I X E C

T I O N

Cipher text: TRIINRI THRSNGTRIEOPTNHEO

- AWDIOPCN

→ Keyword

With keyword = ATLAS 1 2 3 4 5
→ A A L S T

Order them
from left to
right

A T L A S
1 5 3 2 4
t h e a n
n o w i s
p o i n t
i n g n o
R T H d i
R F c t i
O N

1. A: TRPIRRO
2. A: AINNDT
3. L: EWIGHC
4. S: RSTOII
5. T: HOONTEN

Combine:

TRPIRROAINNDTINIGHCRSTOII
HOONTEN

double: Key 1: BREAK
Key 2: LOG

Key 1: 1-A-ainndt
2-B-tapirro
3-E-ewighc
4-K-rstojj
5-R-hoonten

2 5 3 1 4
Key 1: B R E A K
T h e a n
n o w i s
p o i n t
i n g n o
R T H d i
r e c t i
O N

Cipher: Ainndt tapirroewighcrstojjhoonten

Key 2:

2 3 1
L O G 1: G: ntpnwshiot
a i n 2: L: antioictioe
n d t 3: O: idanegnohn
t a p
i a n
o e w
i g h
c a s
t o i
i h o
o n t
c n

ntpnwshiotantioictioeidaneg
nohn.

Decryption:

Ciphertext: ATIDF MRNMF REGOSANOW RSLTP EEFDO
SSSNV AHTOD BIENP GODAE PEZXD ELNSA

Go to

Column

	1	2	3	4	5	6	7	8	9	10
1	T	R	O	O	P	S	H	E	A	D
2	I	N	G	W	E	S	T	N	E	E
3	D	M	O	R	E	S	U	P	P	L
4	I	E	S	S	E	N	D	G	E	N
5	F	R	A	L	D	V	R	O	I	S
6	M	E	N	T	O	A	I	D	X	X

Plaintext: TROOP SHEADING, WEST NEED MORE SUPPL
- ESSEN D GEN F R A L D U B O I S M E N T O A R D X

→ Keyword

SECRET

Ciphertext: ILEDI NHPTK SOJIT MAXSA XERXT
ANONI SNIOT X

total letters = 36

Key length = ~~SECRET~~ = 6

$$36/6 = 6 =$$

I L E D T N

H P T R S O

I I T M A X

S A K E R N

T A N O N I

S N F O T X

Col: = I L E D T N

2 = H P T R S O

3 = I I T M A X

4 = S A K E R N

5 = T A N O N I

6 = S N F O T X

5 2 1 4 3 6

S E C R E T

1 T H I S I S

2 A P L A I N

3 N T E K T F

4 O R D E M O

5 N S T R A T

6 I O N N X X

=> THIS IS A PLAIN TEXT
FOR DEMONSTRATION **

-> Keyword: PLAIN

Cipher: V E X E A X A R A A T X H G D

5 3 1 2 4

P L A I N

H A V E A

G R E A T

D A Y X X

$$15/3 = 3$$

=> HAVE A GREAT DAY **

=> find Inverse permutation of $P(4, 3, 1, 5, 2)$

~~P~~ P 2 3 4 5

Permutation 4 3 1 5 2

P^{-1} 3 5 2 1 4

② 7 4 1 3 5 2 6

1 2 3 4 5 6 7

permutation 7 4 1 3 5 2 6

P^{-1} 3 6 4 2 5 7 1

3) 4 1 3 2 1 2 3 4

P 4 1 3 2

P^{-1} 2 4 3 1

4) 3 5 1 4 2 1 2 3 4 5

P 3 5 1 4 2

P^{-1} 3 5 1 4 2